

A New Didactical Mechanism to Understand Map Projection

M. Elayachi, IAV Hassan2, Rabat, Morocco

INTRODUCTION:

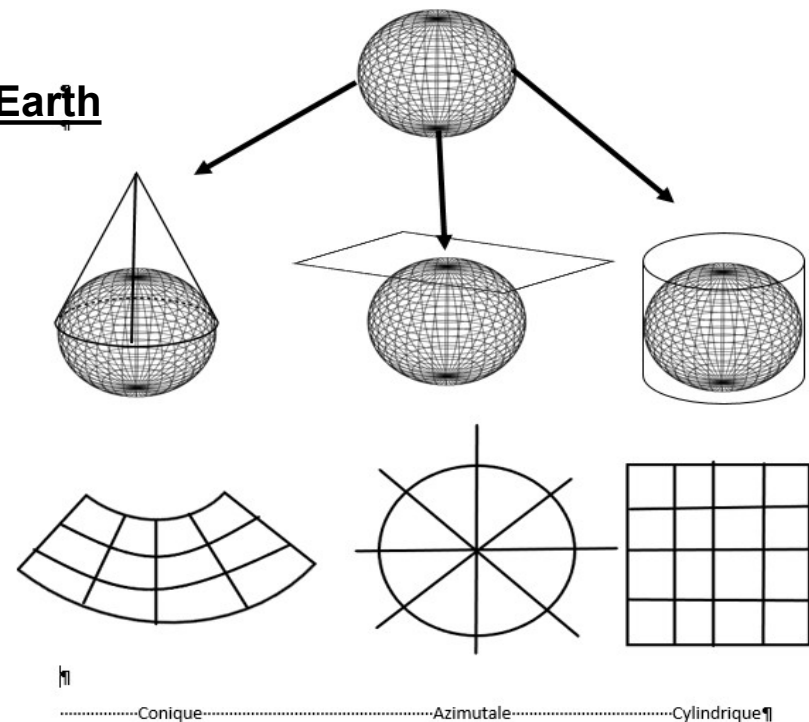
➤ **Projection:** transform the globular 3D shape of the Earth into the 2D form of the map ,

➤ Process: use an intermediate forms :

Conical projections

Cylindrical projections

Azimuthal projections



A New Didactical Mechanism to Understand Map Projection

M. Elayachi, IAV Hassan2, Rabat, Morocco

CHALLENGE :

In Map Projection Process:

- Theoretical approach to teach map projections,
- Complicated steps in terms of mathematics (transformation) and analysis (distorsions)
- Conception needs imagination and theoretical functions

OBJECTIVES :

- Explain practically the phenomenon of projection ?
 - Enable understanding the theoretical steps ?
 - Clarify the transformation process from 3D to 2D representation ?
-

A New Didactical Mechanism to Understand Map Projection

M. Elayachi, IAV Hassan2, Rabat, Morocco

MECHANISM TOOL

✓ Labeling: ISQAT : إسقاط

✓ Component:

- Enlighten globe: Light Bulb (7W, 230 V, 55 mA, 50 Hz)
- A support of the bulb,
- Power cable,
- Grid lines : parallels & meridians
- A transparent sheet.



A New Didactical Mechanism to Understand Map Projection

M. Elayachi, IAV Hassan2, Rabat, Morocco

STEPS

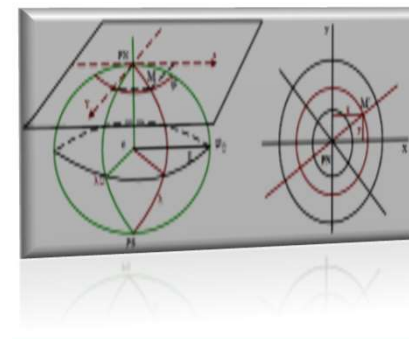
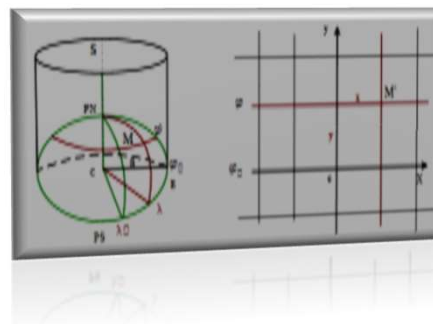
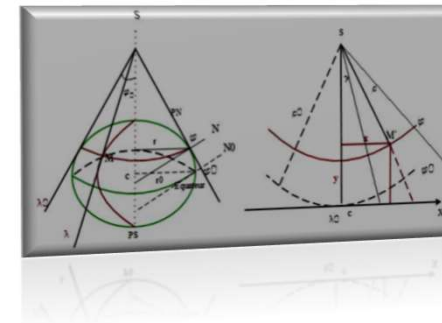
- 1. Use transparent sheet to build : Cylindric, conical , and azimuthal forms*
 - 2. The globe should fit within the three forms,*
 - 3. Draw parallels with a span of 30° : 6 intervalles;*
 - 4. Draw meridians with a span of 45° : 8 intervalles ;*
 - 5. Turn on the globe – ISQAT -*
 - 6. Draw the corresponding images of the grid lines*
-

A New Didactical Mechanism to Understand Map Projection

M. Elayachi, IAV Hassan2, Rabat, Morocco

3. Project the globe -ISQAT- to the plan surface :

- Wrap the globe ISQAT by one transparent surface, cone, cylinder or plane,
- Draw the corresponding images of the grid lines
- Perform the projection by developing the area .



A New Didactical Mechanism to Understand Map Projection

M. Elayachi, IAV Hassan2, Rabat, Morocco

4. Study of the distortions

- Measure a linear feature along a meridian ($S=ds_\phi$) on the globe and its corresponding (S_p) on the projected surface in 3 projections,
- calculate the scale factor S_p/S for each case;
- Measure a linear feature along a parallel ($s=ds_\lambda$) on the globe and its corresponding (S_m) on the projected surface in 3 projections,
- Calculate the factor scale S_m/S for each case;
- calculate the Radius of the globe using the mean radius of the earth (6400 km)
- calculate the factor scale of the transformation from the Earth to the globe (ISQAT).

A New Didactical Mechanism to Understand Map Projection

M. Elayachi, IAV Hassan2, Rabat, Morocco

CONCLUSION

The advantages of the mechanism :

- ✓ Better understanding of the meaning of Map Projections of a 3D form to 2D;
 - ✓ The distortions are concretely identified,
 - ✓ The nature of the obtained image of a projected feature according to the kind of the projection,
 - ✓ Understand the principle of the scale of the transformation;
 - ✓ Save time (6 hours) in teaching the map projections,
 - ✓ This has enabled scheduling the map projections in GIS softwares as added value in the classroom.
-